



Customer Shopping Behavior Analysis

A Data Analytics Project Report

PYTHON

MYSQL

POWER BI

Project at a Glance

3,900

Records Analyzed

18

Dataset Features

\$59.76

Avg. Purchase

3.75

Avg. Review Rating

End-to-end workflow: **Python** cleaning → **MySQL** analysis → **Power BI** dashboard

Dataset Overview

Demographics

Age, Gender, Location,
Subscription

Purchase Details

Item, Category, Amount,
Season, Size, Color

Behavior Signals

Discount, Promo Code, Frequency, Rating, Shipping



Data Cleaning & EDA – Python

→ Loaded & Profiled

3,900 rows, 18 cols — no
duplicates

→ Missing Values

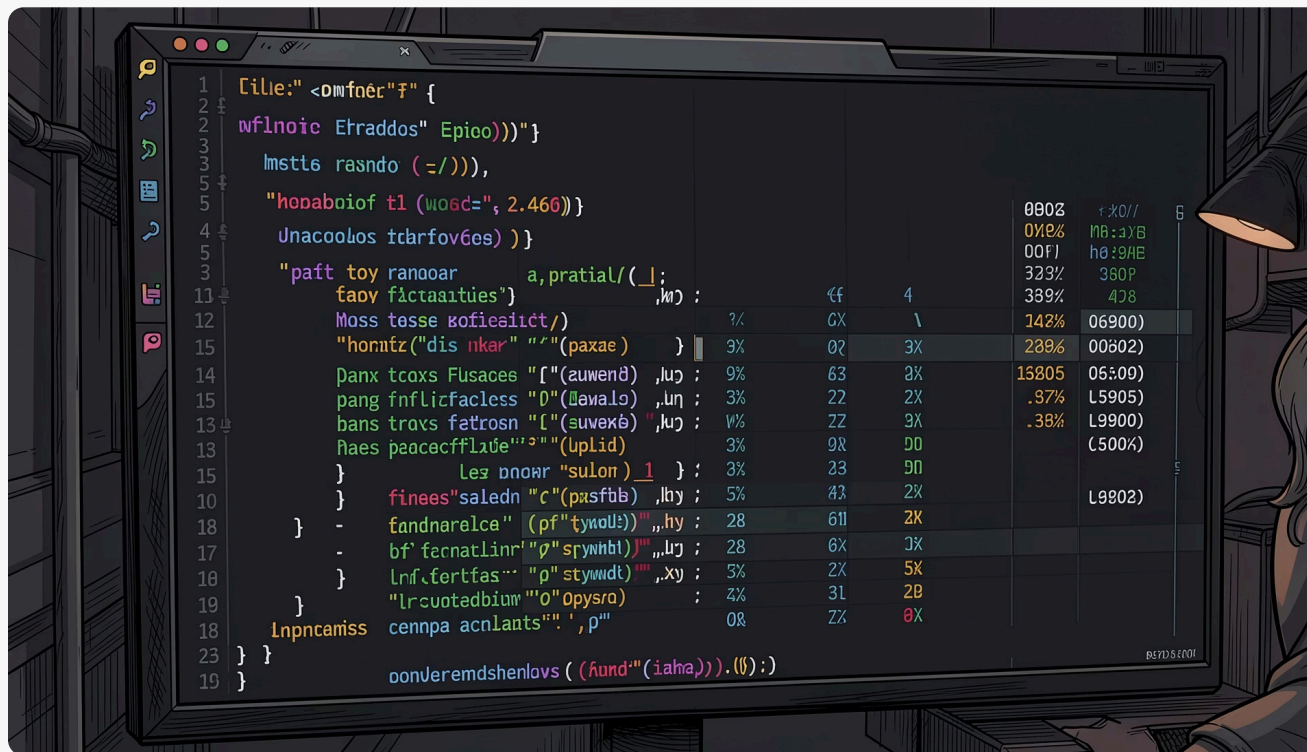
37 nulls in Review Rating →
category-median imputation

→ Feature Engineering

age_group bins +
purchase_frequency_days

→ Redundancy Removed

Promo Code = Discount Applied
→ dropped



SQL Analysis — 10 Business Queries

Subqueries · CTEs · Window Functions · CASE-WHEN · GROUP BY

01

Revenue by Gender

02

High-Spending Discount Users

03

Top 5 Products by Rating

04

Shipping vs. Spend

05

Subscribers vs. Non-Subscribers

06

Discount Rate by Product

07

Customer Segmentation

08

Top 3 Products per Category

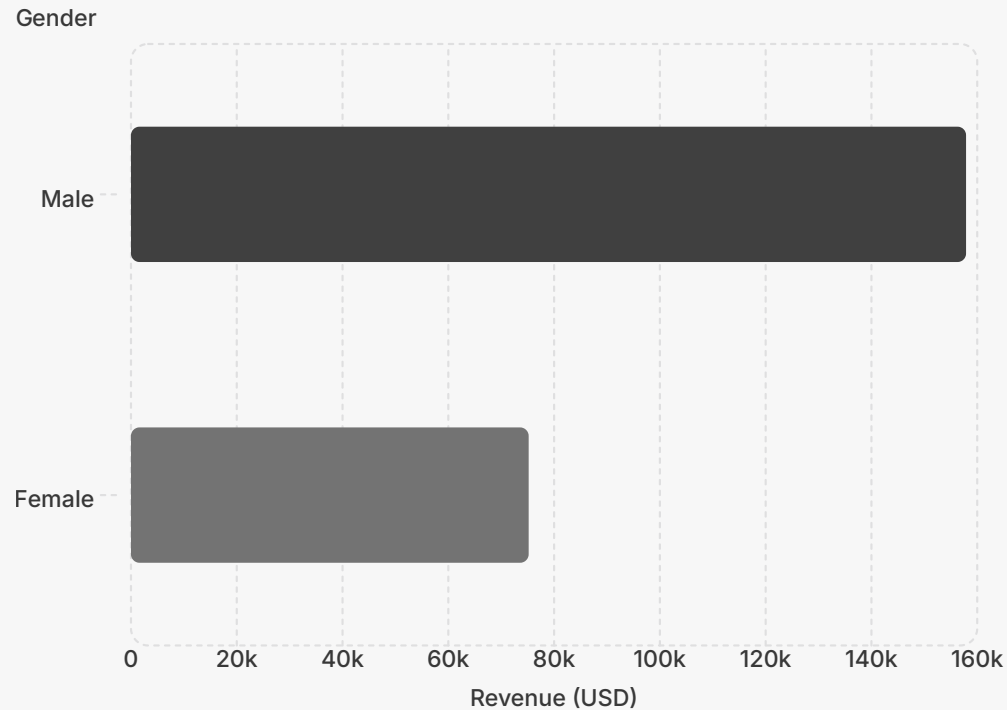
09

Repeat Buyers & Subscription

10

Revenue by Age Group

Key Revenue Insights



Males generate 2x more revenue

\$157,890 vs. \$75,191 — gender-targeted marketing opportunity.

Young Adults lead revenue

\$62,143 — spend is relatively even across age bands.



Subscription Gap — Biggest Opportunity

2,518

Repeat Buyers Not Subscribed

Customers with 5+ purchases — no subscription

958

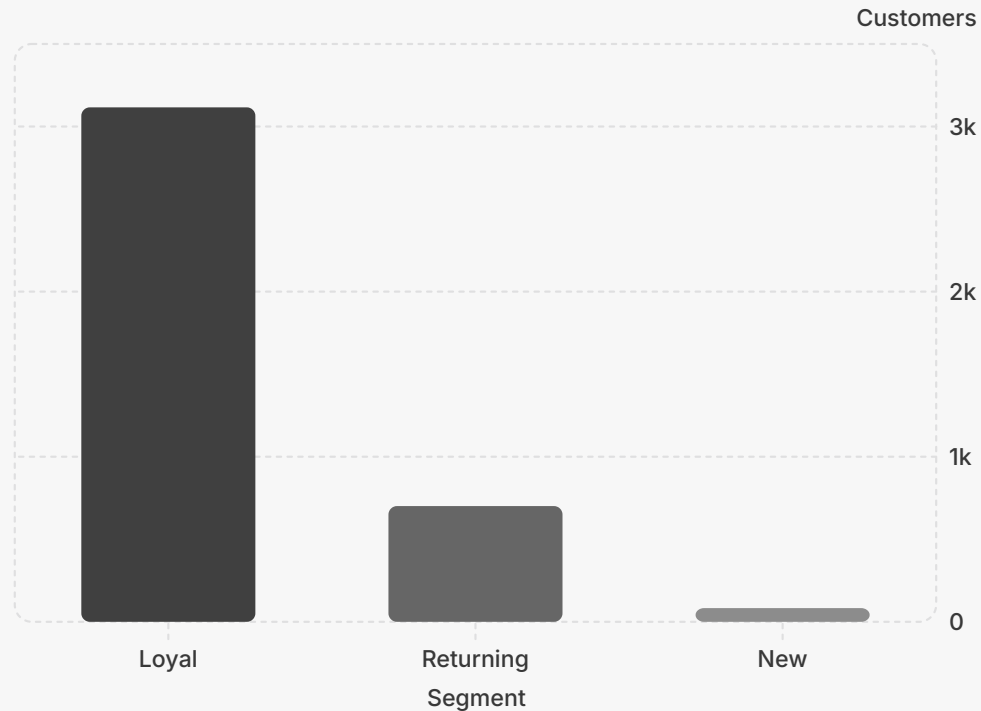
Subscribed Repeat Buyers

Only 27% of loyal customers converted



~\$150K potential recurring revenue untapped — targeted upsell to loyal non-subscribers is the highest-impact action.

Customer Segmentation & Discount Patterns



Loyal customers dominate

80% Loyal, but only 2.1% new — acquisition gap.

Over-discounted products

Hat (50%), Sneakers (49.7%), Coat (49.1%) — margin risk.

839 discount users spent above average

Discounts don't only attract low-value shoppers.

Power BI Dashboard

KPI Cards

3.9K customers · \$59.76 avg ·
3.75 rating

Subscription Split

27% Yes · 73% No

Revenue Charts

By Category & Age Group

Self-Serve Filters

Subscription, Gender, Category,
Shipping



Key Takeaways

Subscription Upsell

2,518 loyal non-subscribers — highest-impact opportunity

Gender Targeting

Males drive 2x revenue — tailor campaigns

Pricing Review

Hats & Sneakers ~50% discount rate — protect margins

Acquisition Gap

Only 83 new customers — invest in growth

